THE GYSIN PULLBACK OF THE DIVISOR OF A MEROMORPHIC SECTION IS THE FIRST CHERN CLASS OF ITS RESTRICTION CAPPED WITH THE FUNDAMENTAL CLASS OF THE DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Local finiteness of a divisor and of its components. Divisors, labels divisors-lemma-components-locally-finite, divisors-lemma-divisor-locally-finite and divisors-lemma-divisor-meromorphic-locally-finite, invoked together at proof lines 5370-5372. Lets the divisor div_N(t) be written as a locally finite sum over integral Z_i of dimension n - 1 with every irreducible component of D among the Z_i, which sets up the coefficient computation.
- The canonical section and the cycle class of a regular section. Divisors, label divisors-definition-regular-meromorphic-ideal-denominators for the canonical section s of O_X(D) (proof lines 5375-5377); same chapter chow.tex, label lemma-compute-c1, invoked at proof line 5400. Gives s with [D]_{n-1} = sum ord_{B_i}(f_i)[Z_i] and identifies div_{L|Z_i}(s_i|Z_i) with [Z_i cap D]_{n-2} for those i with Z_i not contained in D.
- The key formula for the tame symbol. Same chapter chow.tex, label lemma-key-formula, invoked at proof line 5405; the discussion of the section labelled section-key is used at proof line 5378. Supplies the cycle identity that turns the difference of the two coefficient expressions into a sum of divisors of tame symbols, which is exactly the rational equivalence that finishes the proof.

Source reference. Original lemma 0BE1: sources/stacks-chow/tags.txt, source lines 14334–14334. Navigation pointer for \ref{divisors-lemma-components-locally-finite} used inside the counted span (line 5370); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02RL: sources/stacks-chow/tags.txt, source lines 3628–3628. Navigation pointer for \ref{divisors-lemma-divisor-locally-finite} used inside the counted span (line 5371); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SG: sources/stacks-chow/tags.txt, source lines 3659–3659. Navigation pointer for \ref{divisors-lemma-divisor-meromorphic-locally-finite} used inside the counted span (line 5372); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SK: sources/stacks-chow/pinned-original-source.tex, source lines 4198–4198. Navigation pointer for \ref{lemma-compute-c1} used inside the counted span (line 5400); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYC: sources/stacks-chow/pinned-original-source.tex, source lines 4746–4746. Navigation pointer for \ref{lemma-key-formula} used inside the counted span (line 5405); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYB: sources/stacks-chow/pinned-original-source.tex, source lines 4686–4686. Navigation pointer for \ref{section-key} used inside the counted span (line 5378); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5353); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let X be integral and $n = \dim_\delta(X)$. Let $i : D \to X$ be an effective Cartier divisor. Let $\mathcal{N}$ be an invertible $\mathcal{O}_X$-module and let t be a nonzero meromorphic section of $\mathcal{N}$. Then $i^* div_{\mathcal{N}}(t) = c_1(\mathcal{N}|_D) \cap [D]_{n-1}$ in $\mathrm{CH}_{n-2}(D)$.*

Proof. Write $\mathrm{div}_{\mathcal{N}}(t) = \sum \mathrm{ord}_{Z_i,\mathcal{N}}(t)[Z_i]$ for some integral closed subschemes $Z_i \subset X$ of δ-dimension $n-1$. We may assume that the family $\{Z_i\}$ is locally finite, that $t \in \Gamma(U, \mathcal{N}|_U)$ is a generator where $U = X \setminus \bigcup Z_i$, and that every irreducible component of D is one of the Z_i, see Divisors, Lemmas 0BE1, 02RL, and 02SG.

Set $\mathcal{L} = \mathcal{O}_X(D)$. Denote $s \in \Gamma(X, \mathcal{O}_X(D)) = \Gamma(X, \mathcal{L})$ the canonical section. We will apply the discussion of Section 0AYB to our current situation. For each i let $\xi_i \in Z_i$ be its generic point. Let $B_i = \mathcal{O}_{X,\xi_i}$. For each i we pick generators $s_i \in \mathcal{L}_{\xi_i}$ and $t_i \in \mathcal{N}_{\xi_i}$ over B_i but we insist that we pick $s_i = s$ if $Z_i \not\subset D$. Write $s = f_i s_i$ and $t = g_i t_i$ with $f_i, g_i \in B_i$. Then $\mathrm{ord}_{Z_i,\mathcal{N}}(t) = \mathrm{ord}_{B_i}(g_i)$. On the other hand, we have $f_i \in B_i$ and

$$[D]_{n-1} = \sum \mathrm{ord}_{B_i}(f_i)[Z_i]$$

because of our choices of s_i. We claim that

$$i^*\mathrm{div}_{\mathcal{N}}(t) = \sum \mathrm{ord}_{B_i}(g_i)\mathrm{div}_{\mathcal{L}|_{Z_i}}(s_i|_{Z_i})$$

as cycles. More precisely, the right hand side is a cycle representing the left hand side. Namely, this is clear by our formula for $\mathrm{div}_{\mathcal{N}}(t)$ and the fact that $\mathrm{div}_{\mathcal{L}|_{Z_i}}(s_i|_{Z_i}) = [Z(s_i|_{Z_i})]_{n-2} = [Z_i \cap D]_{n-2}$ when $Z_i \not\subset D$ because in that case $s_i|_{Z_i} = s|_{Z_i}$ is a regular section, see Lemma 02SK. Similarly,

$$c_1(\mathcal{N}) \cap [D]_{n-1} = \sum \mathrm{ord}_{B_i}(f_i)\mathrm{div}_{\mathcal{N}|_{Z_i}}(t_i|_{Z_i})$$

The key formula (Lemma 0AYC) gives the equality

$$\sum \Big(\mathrm{ord}_{B_i}(f_i)\mathrm{div}_{\mathcal{N}|_{Z_i}}(t_i|_{Z_i}) - \mathrm{ord}_{B_i}(g_i)\mathrm{div}_{\mathcal{L}|_{Z_i}}(s_i|_{Z_i})\Big) = \sum \mathrm{div}_{Z_i}(\partial_{B_i}(f_i, g_i))$$

of cycles. If $Z_i \not\subset D$, then $f_i = 1$ and hence $\mathrm{div}_{Z_i}(\partial_{B_i}(f_i, g_i)) = 0$. Thus we get a rational equivalence between our specific cycles representing $i^*\mathrm{div}_{\mathcal{N}}(t)$ and $c_1(\mathcal{N}) \cap [D]_{n-1}$ on D. This finishes the proof. □